Peer-graded Assignment: Week 4: Non-parametric Models

Reviews 2 left to complete

Your fellow learner has submitted their assignment anonymously and your review will be anonymous to them. All names are still visible to course instructors.

Module4_v4

by Anonymous Learner
October 19, 2023

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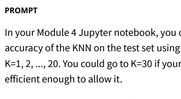
PROMPT

For Problem 1 of the Module 4 lab, you created a KNN classifier for handwritten digit recognition. Use your KNN class to perform KNN on the validation data with K = 3. Now you will check how well your classifier does. Create a **confusion matrix** in your Jupyter notebook. Feel free to use the Scikit-Learn confusion matrix (http://scikit-learn.org/stable/modules/generated/sklearn.metrics.confusion_matrix.html) function).

Upload a screenshot of the confusion matrix from your notebook here.



RUBRIC

Did the author upload a screenshot of a **confusion matrix** showing the performance of their KNN classifier on the 9 digits in the MNIST dataset?

See an example solution below:

```
confusion matrix:
[[112   0   0   0   0   0   1   0   0   0]
 [  0 106   0   0   0   0   0   1   0   1]
 [  1   2  86   2   0   0   0   2   0   0]
 [  1   1   0 111   0   2   0   0   0   0]
 [  0   2   0   0  82   0   0   0   0   4]
 [  0   0   0   2   2  75   1   0   0   0]
 [  0   0   0   0   0   1 104   0   2   0]
 [  0   2   0   0   0   1   0  93   0   5]
 [  1   1   0   1   1   0   2   1  81   1]
 [  1   0   0   1   2   0   0   2   0 100]]
```

- 2 pts
  Learner's response looks reasonable for a confusion matrix showing the performance of their KNN classifier on the 9 digits in the MNIST dataset, e.g. they submitted a **10 x 10 confusion matrix** .
- 0 pts
  Learner's response does not look reasonable for a confusion matrix showing the performance of their KNN classifier on the 9 digits in the MNIST dataset, e.g. they **didn't submit a 10 x 10 confusion matrix .**

PROMPT

Based on your confusion matrix, which digits seem to get confused with other digits the most?

Type your answer in the rich text field.

Digits 9 and 7 seem to be confused with other digits for having the most off-diagonal entries in the same row/column.

RUBRIC

Does the author correctly identify which digits are confused with other digits the most?

- 2 pts
  Correctly identifies that 9 and 7 are confused with other digits the most.

  See how the confusion matrix shows that these two digits were misclassified five times.

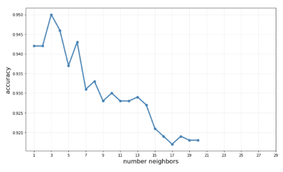
```
confusion matrix:
[[112   0   0   0   0   0   1   0   0   0]
 [  0 106   0   0   0   0   0   1   0   1]
 [  1   2  86   2   0   0   0   2   0   0]
 [  1   1   0 111   0   2   0   0   0   0]
 [  0   2   0   0  82   0   0   0   0   4]
 [  0   0   0   2   2  75   1   0   0   0]
 [  0   0   0   0   0   1 104   0   2   0]
 [  0   2   0   0   0   1   0  93   0   5]
 [  1   1   0   1   1   0   2   1  81   1]
 [  1   0   0   1   2   0   0   2   0 100]]
```

- 1 pt
  Does not correctly identify that 9 and 7 are confused with other digits the most, but the digits identified are correct based on the confusion matrix that the learner uploaded.
- 0 pts
  Learner did not answer the question **or** learner did not identify that 9 and 7 are confused with other digits the most **and** digits the learner identified were not correct based on the confusion matrix that the learner uploaded.

PROMPT

In your Module 4 Jupyter notebook, you created a plot of the accuracy of the KNN on the test set using the same set of axes for K=1, 2, .., 20. You could go to K=30 if your implementation was efficient enough to allow it.

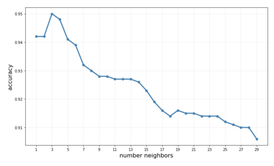
Please upload a **screenshot of your plot** here**.**



RUBRIC

Evaluate the learner's KNN accuracy plot.

Here is a sample solution plot:



- 4 pts
  Plot has K values on the x-axis **and** test set accuracy on the y-axis
- 2 pts
  Plot **missing at least one of the following**: K values on x-axis **or** test set accuracy on the y-axis.
- 0 pts
  No plot submitted.

PROMPT

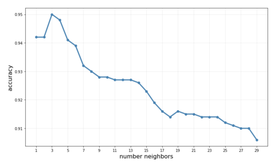
Based on the plot of the KNN accuracy on the test set, which value of K results in highest accuracy?

Type your answer in the rich text field.

The accuracy peaks at K = 3 according to the figure.

RUBRIC

From the solution KNN accuracy plot below, we can see that a K value of 3 neighbors results in the highest accuracy.



- 4 pts
  Learner correctly identifies that 3 neighbors results in the highest accuracy .
- 2 pts
  Learner does not identify a K value of 3 but identifies the highest accuracy K value on the plot that the learner submitted for Prompt 3.
- 0 pts
  No answer attempted **or** learner does not identify highest accuracy K value on the plot that the learner submitted.

PROMPT

In Problem 2 about Decision Trees, the directions for Part C1 are to:

1. Modify **max_depth**:
   - Create a model with a shallow **max_depth** of 2. Build the model on the training set with the **build_dt** function.
   - Report **precision/recall** on the test set.
   - Report **depth of the tree**.

Upload a screenshot of your code for this section.

RUBRIC

Here is a **solution**:

```
dt = build_dt(X_train, y_train, max_depth= 2)


y_test_hat = dt.predict(X_test)


prediction = calculate_precision(y_test, y_test_hat)


recall = calculate_recall(y_test, y_test_hat)

print("Precision: {:0.2f} Recall: {:0.2f} Tree Depth: {}".format(prediction, recall, dt.get_depth()))
```

Evaluate the learner's submission.

- 5 pts
  Includes all of the following: Learner calls build_dt function with max_depth=2 **and** model built on the training set **and** calculates precision on the test set **and** calculates recall on the test set **and** reports depth of the tree.
- 4 pts
  Missing one of the following: Learner calls build_dt function with max_depth=2 **or** model built on the training set or calculates precision on the test set **or** calculates recall on the test set **or** reports depth of the tree.
- 3 pts
  Missing two of the following: Learner calls build_dt function with max_depth=2 **or** model built on the training set or calculates precision on the test set **or** calculates recall on the test set **or** reports depth of the tree.
- 2 pts
  Missing three of the following: Learner calls build_dt function with max_depth=2 **or** model built on the training set or calculates precision on the test set **or** calculates recall on the test set **or** reports depth of the tree.
- 1 pt
  Missing four of the following: Learner calls build_dt function with max_depth=2 **or** model built on the training set or calculates precision on the test set **or** calculates recall on the test set **or** reports depth of the tree.
- 0 pts
  Missing all of the following: Learner calls build_dt function with max_depth=2 **and** model built on the training set **and** calculates precision on the test set **and** calculates recall on the test set **and** reports depth of the tree.

PROMPT

In Problem 2 about Decision Trees, the directions for Part C1 are to:

2. Modifying **max_leaf_nodes**:
   - Create a model with a shallow **max_leaf_nodes of 4**. Build the model on the training set.

Upload a screenshot of your code for this section.

RUBRIC

Here is a solution:

```
13
```

Evaluate the learner's solution.

- 5 pts
  Includes all of the following: Learner calls build_dt function with max_leaf_nodes = 4 **and** model built on the training set